

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** STOCK TRADING APP (SB Stocks)
- **Team Members:** List team members and their roles.
 - U. Issak – Developer & Tester
 - Team Member 2 – Frontend Developer
 - Team Member 3 – Backend Developer
 - Team Member 4 – UI/UX Designer

2. Project Overview

- **Purpose:** To build a paper trading web application that provides real-time US stock market data, allowing users to buy/sell virtual stocks, manage portfolio, and track performance using virtual funds.
- **Features:** Real-time stock prices, user authentication, buy/sell stocks, portfolio dashboard, transaction history, watchlist, profile management, and virtual balance.

3. Architecture

- **Frontend:** React.js (Vite), Bootstrap, React Router, Axios, Chart.js
- **Backend:** Node.js, Express.js, RESTful APIs, JWT Authentication
- **Database:** MongoDB (Mongoose ODM) – Stores users, stocks, portfolios, orders, transactions, and watchlist data.

4. Setup Instructions

- **Prerequisites:** Node.js, MongoDB, Git
- **Installation:** Step-by-step guide to clone, install dependencies, and set up the environment variables.
 - Clone the repository
 - Install dependencies for client and server
 - Configure .env file for server
 - Start MongoDB service

5. Folder Structure

- **Client:** React frontend (components, pages, assets, services, context, etc.)
- **Server:** Node.js backend (controllers, models, routes, middleware, config, etc.)

6. Running the Application

- Provide commands to start the frontend and backend servers locally.
 - **Frontend:** npm run dev (or npm start) in the client directory.
 - **Backend:** npm start in the server directory.

7. API Documentation

- Document all endpoints exposed by the backend.
- Include request methods, parameters, and example responses.
(e.g., /api/auth/register, /api/auth/login, /api/stocks, /api/orders/buy, /api/portfolio, etc.)

8. Authentication

- Explain how authentication and authorization are handled in the project.
- JWT (JSON Web Token) is used for authentication.
- Protected routes require a valid token in Authorization header.

9. User Interface

- Provide screenshots or GIFs showcasing different UI features.
(Home Page, Stocks Page, Portfolio Dashboard, Buy/Sell Modal, Transaction History, Profile Page)

10. Testing

- Describe the testing strategy and tools used.
(Manual Testing, Thunder Client for API testing, Browser Testing)

11. Screenshots or Demo

- Provide screenshots or a link to a demo to showcase the application.
Demo Link: <http://localhost:5174>

12. Known Issues

- Document any known bugs or issues that users or developers should be aware of.
(e.g., Real-time price delay in some cases, UI responsiveness on smaller screens, etc.)

13. Future Enhancements

- Outline potential future features or improvements that could be made to the project.
(Real-time WebSocket updates, Advanced charts, Multiple watchlists, Email notifications, Admin panel, Deploy to cloud, etc.)